

Kaiming Liu

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Research Profile Machine learning for probabilistic inference and robust prediction in scientific inverse problems with multiple plausible solutions.

Education

University of Illinois Urbana-Champaign
Ph.D. in Physics, in progress Aug 2023–present
Advisor: Bryan K. Clark

Xi'an Jiaotong University
B.S. in Physics (Honors) Sep 2019–Jul 2023
Special Class for the Gifted Young Sep 2017–Jul 2019

University of California, Berkeley
BPIE Program (non-degree visiting program) Aug–Dec 2021

Current Research **University of Illinois Urbana-Champaign**
Graduate researcher, Bryan K. Clark group Nov 2025–present

Conditional density modeling / crystal-field parameters

- Built residual-MLP conditional Gaussian mixtures for conditional density estimation from magnetic susceptibility.
- Represented multiple plausible solutions and checked them by reconstructing susceptibility; studied robustness on simulated and experimental data.
- Evaluated conditional Gaussian mixtures and conditional RealNVP normalizing flows in a pilot density-model comparison.

Robust inverse prediction / neutron scattering

- Trained MLP and ResNet conditional Gaussian mixtures to infer exchange parameters from high-dimensional neutron-scattering maps.
- Checked predictions with Sunny forward simulations and classical fit references; tested noise, intensity scaling, and background offsets.

Neural-network backflow / JAX and Monte Carlo

- Implemented neural-network backflow and variational Monte Carlo in JAX for Hubbard models, with GPU/multi-GPU sampling.
- Profiled local-energy evaluation and gradient synchronization; verified chunked evaluation and global reductions against reference computations.

Publications	[P1] Evidence of intertwined pair density and charge density wave orders in UTe₂ Zhen Zhu, Yudi Huang, Julian May-Mann, Kaiming Liu, Zheyu Wu, Shanta R. Saha, Johnpierre Paglione, Alexander G. Eaton, Andrej Cabala, Michal Vališka, Eduardo Fradkin, Vidya Madhavan. <i>PNAS</i> 123 (29), e2602117123 (2026). arXiv.	
Prior Research	University of Illinois Urbana-Champaign <i>Research assistant, Vidya Madhavan group</i> Aug 2023–Oct 2025 • Grew thin-film heterostructures with MBE and characterized electronic structure with STM/STS in ultrahigh vacuum. • Automated growth temperature control and built Python/MATLAB tools for quasiparticle-interference analysis. Xi'an Jiaotong University <i>Research assistant, Yongchang Zhang group</i> Mar 2022–Jul 2023 • Simulated Rydberg-atom dynamics and photon-mediated interactions in Python/MATLAB to study quantum nonlinear optics. • Built a discrete-time lattice model of two-qubit phase-flip dynamics and many-body regimes. University of California, Berkeley <i>Research assistant, Michael F. Crommie group</i> Oct–Dec 2021 • Assisted synthesis and STM characterization of graphene nanoribbons with five-membered rings to study their electronic states.	
Teaching	University of Illinois Urbana-Champaign <i>Teaching assistant, PHYS 212</i> Spring & Fall 2026 <i>Teaching assistant, PHYS 102</i> Aug–Dec 2023	
Selected Honors	China National Scholarship 2021 Xi'an Jiaotong University Mathematical Contest in Modeling, Meritorious Winner 2021 COMAP	
Technical Skills	<i>ML / modeling</i>: PyTorch, JAX; conditional mixtures, residual networks, probabilistic inference. <i>Research engineering</i>: Python, NumPy, SciPy, Git, Linux; GPU/multi-GPU and cluster computation, runtime profiling. <i>Numerical / scientific methods</i>: MCMC, VMC, numerical optimization; forward-model evaluation and simulation-to-experiment robustness. <i>Experimental physics</i>: MBE, STM/STS, UHV; feedback control, automation, MATLAB and data analysis.	